

Will Steinhardt

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Appointments

UC Santa Cruz | Assistant Researcher

UC Santa Cruz | Postdoctoral Scholar

2025-Current

2020-2025

Education

Harvard University | Ph.D. in Earth & Planetary Sciences

Caltech | B.S. in Geophysics

Feb 2020

2011

Honors and Awards

Jason Morgan Early Career Award | AGU

Koret Scholar Mentoring Award | UCSC

Outstanding Student Presentation Award | AGU

Shaler Teaching Award | Harvard Earth & Planetary Sciences

Certificate of Distinction in Teaching | Harvard Bok Center

NASA Earth and Space Science Fellowship | NASA (declined)

Fritz Burns Prize | Caltech Division of Geologic & Planetary Sciences

Student Life and Masters Award | Caltech

2024

2023

2018

2016

Spring 2015, Spring 2016

2013

2011

2011

Publications

1. Steinhardt, W. & Brodsky, E. E. "Distinguishing Single and Linked Ruptures in the Laboratory and Nature" (Accepted, *GRL*)
2. Steinhardt, W. & Brodsky, E. E. "Precursory Locking Precedes Slip Events on Laboratory Fault". *Geophysical Research Letters* 52, e2024GL112882 (2025). ([LINK](#))
3. **Steinhardt, W.** & Rubinstein, S. M. "Geometric rules for the annihilation dynamics of step lines on fracture fronts". *Phys. Rev. E* 107, 055003 (2023). ([LINK](#))
4. **Steinhardt, W.**, Dillavou, S., Agajanian, M., Rubinstein, S. M. & Brodsky, E. E. "Seismological Stress Drops for Confined Ruptures Are Invariant to Normal Stress". *Geophysical Research Letters* 50, e2022GL101366 (2023). ([LINK](#))
5. **Steinhardt, W.** & Rubinstein, S.M. "How Material Heterogeneity Creates Rough Fractures" *Phys. Rev. Lett.*, 129, 128001 (2022). ([LINK](#)) ([Chosen For Cover](#))
6. Solomatova, N.V., Jackson, J.M., Sturhahn, W., Wicks, J.K., Zhao, J., Toellner, T.S., Kalkan, B. and **Steinhardt, W.** "Equation of state and spin crossover of (Mg, Fe) O at high pressure, with implications for explaining topographic relief at the core-mantle boundary." *American Mineralogist*, 101(5), pp.1084-1093. (2016).
7. Bindi, L., Yao, N., Lin, C., Hollister, L.S., Andronicos, C.L., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., MacPherson, G.J., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., "Decagonite, Al₇₁Ni₂₄Fe₅, a quasicrystal with decagonal symmetry from the Khatyrka CV3 carbonaceous chondrite." *American Mineralogist*, 100(10), pp.2340-2343. (2015).
8. Bindi, L., Yao, N., Lin, C., Hollister, L.S., Andronicos, C.L., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., MacPherson, G.J., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., 2015. "Natural quasicrystal with decagonal symmetry." *Scientific Reports*, 5, p.9111. (2015).
9. Bindi, L., Yao, N., Lin, C., Hollister, L.S., MacPherson, G.J., Poirier, G.R., Andronicos, C.L., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., "Steinhardtite, a new body-centered-cubic allotropic form of aluminum from the Khatyrka CV3 carbonaceous chondrite." *American Mineralogist*, 99(11-12), pp.2433-2436. (2014).
10. Hollister, L.S., Bindi, L., Yao, N., Poirier, G.R., Andronicos, C.L., MacPherson, G.J., Lin, C., Distler, V.V., Eddy, M.P., Kostin, A., Kryachko, V., **Steinhardt, W.**, Yudovskaya M., Eiler J.M., Guan Y., Clarke J.J., Steinhardt P.J., "Impact-induced shock and the formation of natural quasicrystals in the early solar system." *Nature Communications*, 5, p.4040. (2014).
11. MacPherson, G.J., Andronicos, C.L., Bindi, L., Distler, V.V., Eddy, M.P., Eiler, J.M., Guan, Y., Hollister, L.S., Kostin, A., Kryachko, V., **Steinhardt, W.**, Yudovskaya M., Steinhardt P.J., "Khatyrka, a new CV 3 find from the Koryak Mountains, Eastern Russia." *Meteoritics & Planetary Science*, 48(8), pp.1499-1514. (2013).
12. M. Obrić, Ž Ivezić, P. N. Best, R. H. Lupton, C. Tremonti, J. Brinchmann, M. A. Agüeros, G. R. Knapp, J. E. Gunn, C. M. Rockosi, D. Schlegel, D. Finkbeiner, M. Gaćša, V. Smolčić, S. F. Anderson, W. Voges, M. Jurić, R. J. Siverd, **Steinhardt, W.**, A. S. Jagoda, M. R. Blanton, D. P. Schneider; "Panchromatic properties of 99 000 galaxies detected by SDSS, and (some by) ROSAT, GALEX, 2MASS, IRAS, GB6, FIRST, NVSS and

WENSS surveys”, Monthly Notices of the Royal Astronomical Society, Volume 370, Issue 4, 1677–1698 (2006).

13. Sesar, B., Svilković, D., Ivezić, Ž., Lupton, R.H., Munn, J.A., Finkbeiner, D., **Steinhardt, W.**, Siverd, R., Johnston, D.E., Knapp, G.R. and Gunn, J.E., “Variable faint optical sources discovered by comparing the POSS and SDSS catalogs.” The Astronomical Journal, 131(6), p.2801. (2006).
14. Ivezić, Ž., Richards, G., Hall, P., Lupton, R., Jagoda, A., Knapp, G., Gunn, J., Strauss, M., Schlegel, D., **Steinhardt, W.** and Siverd, R., June. “Quasar Radio Dichotomy: Two Peaks, or not Two Peaks, that is the Question.” AGN Physics with the Sloan Digital Sky Survey (Vol. 311, p. 347). (2004).

Invited Talks

Gordon Research Conference (Rock Mechanics)	8 / 2026
Gordon Research Conference (Soft Matter)	7 / 2026
USGS Earthquake Science Seminar	4 / 2026
KITP Soft Matter Geophysics Seminar	1 / 2026
Berkeley Earth Science Seminar	1 / 2025
Cornell Civil and Environmental Engineering Seminar	12 / 2024
Berkeley Seismo Lab Seminar	10 / 2024
Princeton Center for Theoretical Physics: Fracture Across Fields	5 / 2024
Computational Infrastructure for Geodynamics Seminar	3 / 2023
Brandeis MRSEC Seminar	10 / 2022
MIT Geophysics Seminar	10 / 2022
USGS Earthquake Science Center Seminar	7 / 2022
University of Washington Earth and Space Science Colloquium	1 / 2022
UCSC IGPP Seminar	2 / 2020
American Physical Society (March Meeting 2019)	3 / 2019

Teaching

Teaching Fellow for <i>ENG-SCI 123: Introduction to Fluid Mechanics and Transport Processes</i>	Spr 2015, Spr 2016
Teaching Assistant for <i>GE 1: Earth and Environment</i>	Spr 2010

Service and Outreach

Reviewer for <i>Scientific Reports</i>	
Reviewer for <i>Review of Scientific Instruments</i>	
Reviewer for <i>Journal of Geophysical Research</i>	
Organized UCSC IGPP Seminar	2020 - 2022
Supervised undergraduate Shannon Iron on earthquake statistics research	2023 - Present
Supervised undergraduate Jaiden Zak on friction research	2022 - 2023
Supervised undergraduate Sydney Haith on interacting asperity research	2022 – 2024
Supervised graduate student Caroline Martin on fracture surface analysis project	2019 - 2019
Supervised graduate student Rodrigo Telles on hydrogel heterogeneity project	2018 - 2019
Supervised undergraduate Aria Hamann on interfacial hydrogel fracture project through Harvard REU Program	2014
Led <i>Science Days</i> lab tours and demonstrations for middle school students	2015, 2016